

- A. Outline the anaesthetic concerns associated with elective endovascular repair of infra-renal abdominal aortic aneurysm (EVAR) (50%).**
- B. Discuss the factors favouring monitored anaesthesia care (MAC) as the choice of anaesthesia for EVAR (50%).**

A.

Anaesthetic concerns include:

#### **Patient**

Ability to cooperate/ lie supine/ breath hold

- Affects suitability for MAC if patient is unable to cooperate.
- GA may be necessary to facilitate optimal operating conditions.

Risks of airway obstruction/ difficult airway

- MAC may not be suitable if patient is assessed to have high risk of airway obstruction

Cardiovascular co-morbidities

- AAA associated with coronary artery disease.
- Greater haemodynamic instability with GA if patient is elderly with poor cardiovascular reserves
- Invasive lines (IA, CVP) for monitoring and vasopressors may be necessary for safe conduct of GA.
- Starting Hb should be optimised, with valid GXM.

Risk factors for acute kidney injury (AKI)

- Pre-existing renal impairment, use of nephrotoxic drugs (NSAIDs, antibiotics), dehydration increases risk of peri-operative AKI with contrast use.
- Preventive measures such as pre-op sodium bicarbonate infusion, N-acetylcysteine use, avoid further nephrotoxic drugs, maintain preload, and peri-op monitoring of renal function may reduce risk of AKI.

Management of anti-platelets/ anti-coagulants (if any)

- Anti-platelets are usually continued, which also help with prevention of stent thrombosis.
- Indications for anti-coagulants should be reviewed, and need for cessation should be discussed pre-op.

#### **Surgical**

Complexity of procedure

- Previous graft in-situ, presence of endoleak, tortuous aneurysm, large sized >7cm, involvement of renal/ common iliac arteries, long segment aneurysm can make procedure more complex.
- Choice of stent/graft also affects complexity of procedure.
  - o Type (chimney/ bare metal/ fenestrated)
  - o Length
  - o Number of stent-grafts required
  - o Availability of suitable landing site for vascular stent/grafts
- This affects duration of surgery, bleeding/ thrombotic complications, and also risk of conversion to open repair.

Vascular approach and access required

- Usually femoral access, but subclavian/ brachial access may be required if a top-down approach is used (e.g. diseased LL arteries).
- This limits location for radial IA cannulation.

Use of iodinated radio-contrast

- Associated with peri-op AKI.
- Risk increases with higher doses e.g. for longer/ complex cases
- Occupational hazard to staff

- Use of lead gown/ shield
  - Pregnant staff should not be involved in the case
- CO2 contrast may be used as an alternative in high risk/ susceptible patients.
  - Risk of gas embolism.
  - Accumulation can lead to hypercarbia.
  - Ischaemic pain.
  - Patient may need to breath-hold in order to obtain decent images.

#### Blood loss

- Should be discussed with surgeon, especially if anticipated complex surgery with bleeding risks.
- Valid GXM is necessary.
- Cell salvage not feasible with endovascular technique.
- Bleeding points/ inadvertent vessel injury may be occult and may not be immediately recognizable.
- Risk of aneurysm rupture highest during stent deployment.
- Close communication between surgeon and anaesthetist, with increased vigilance during this phase, will facilitate early detection and treatment

#### Risk of spinal cord injury

- Low but not negligible.
- Especially if lumbar arteries embolization is done pre-op
- Close neurological monitoring needed to facilitate early detection and intervention.

#### Requirement for Image Intensifier (II)

- Will need to ensure availability of hybrid OT with II support

#### Risk of conversion to open procedure

- Remote location at IR suite may not be appropriate for EVAR.
- Hybrid OT will be more appropriate, especially if risk of conversion is high. Cross clamp of aorta will be needed, increasing risks of distal ischaemia below level of clamp.

### **Anaesthesia**

#### Choice of anaesthesia

- Patients listed for EVAR (instead of open repair) are often older with more significant comorbidities, and may not be good candidates for GA.
- Spinal anaesthesia is appropriate, and will require peri-op cessation of anti-platelet/ anti-coagulants.
- MAC confers greater haemodynamic stability, which will need to be weighed against patient's ability to cooperate, risks of airway obstruction, and duration of surgery.

#### Impact of haemodynamic swings on transmural pressure on aneurysmal vessel wall

- Increases risks of aneurysm rupture, which can be catastrophic.
- IA monitoring is essential
- Careful BP control to avoid hypertension/ hypotension/ large BP swings is critical to minimise this risk.

#### Management of pain/ sympathetic surges

- Aside from pain arising from puncture site, ischaemic pain from LL may occur during long procedures due to occlusion of femoral/ iliac vessels. This needs to be managed with analgesia top up.
- Spinal anaesthesia confers superior analgesia compared to systemic analgesia or local infiltration of LA.

B.

EVAR is a procedure where there is minimal pain stimulus following initial femoral puncture, which makes MAC an appropriate choice of anaesthesia. However, it is important to ensure availability of airway equipment, resuscitation trolley, and blood products in the event of deterioration/ complications requiring conversion to GA.

**MAC is the choice of anaesthesia for EVAR due to the following benefits**

- Better haemodynamic profile
- Avoids need for invasive airway (and associated potential complications)
- Avoids risk of spinal haematoma with CNB approach, especially with intra-op heparinization
- Facilitates neuro-monitoring
- Quicker OT turnaround time
- Improved patient recovery post-op

There are also patient/ surgical/ anaesthesia factors which will impact the suitability of using MAC for EVAR.

**Patient**

Ability of patient to cooperate/ lie supine/ breath hold

- Calm and cooperative patient facilitates optimal operating conditions for EVAR under MAC
- Precludes patients who are anxious/ cognitively impaired, especially if breath holding is required during certain phases (stent deployment for juxta-renal AAA).

Low risks of airway obstruction/ difficult airway

Ability to maintain spontaneous ventilation with no underlying respiratory issues

- This confers greater margin of safety for maintenance of airway patency and respiratory drive when titrating sedative agents.

Presence of significant cardiovascular co-morbidities

- MAC confers greater haemodynamic instability, especially with slow controlled titration, as compared to GA.

No significant kidney impairment/ failure

- Use of CO2 contrast (in lieu of iodinated radio-contrast) may be associated with more pain and may not be well tolerated under MAC.

**Surgical**

Experienced Surgeon

Low procedural complexity (Simple infra-renal AAA)

- Lower risk of conversion to open, which would necessitate GA.
- Lower risks of massive blood transfusion, which may necessitate securing the airway.

Short op duration (less than 2 hours)

- This will lead to higher success rates as longer procedures may not be as well-tolerated under MAC.
- Longer procedures may be associated with possible re-perfusion syndrome. Treatment and resuscitation for this is better done under GA with secured airway.

**Anaesthesia**

Experienced Anaesthetist

- Good working relationship with vascular surgeon
- Familiar with the procedural steps and potential complications
- Comfortable with optimising sedation and analgesia for patient comfort and operating conditions

**28.6% candidates passed this question (6/21 passed)**

Candidates performed poorly in this question. A significant number of them misinterpreted the question and only focused on describing the pros and cons of MAC for EVAR. Those that passed were able to mention the basic requirements for patients to undergo EVAR under MAC for example (a) simple infrarenal AAA that will not take long; (b) cooperate patients that can remain still during the operation and (c) surgeons that can tolerate minor movements of patients. Better candidates can show their understanding of the requirements the possibilities of patients to stopping breathing for some time during juxtarenal AAA or discuss on why MAC is not a good choice for long procedures in view of possible re-perfusion syndrome, to name a few.